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Digital Transformation at Tata Steel

May 2022: T.V. Narendran, CEO of Tata Steel, India's oldest steel manufacturer, was considering the firm's future.¹ Since 2013, as CEO, he had taken concrete business and cultural transformative steps to "future-ready" the firm by deleveraging and instilling financial discipline, acquiring new companies (within steel), entering new segments and adjacent businesses, and launching digital transformation, agility, safety, and sustainability programs.

These measures had transformed Tata Steel into an advanced steel manufacturer globally, with its Dutch plant being the first steel manufacturing unit in the world to be assessed as a World Economic Forum (WEF) Lighthouse in 2018.^{a,2} Soon after, the firm's Kalinganagar and Jamshedpur plants also received the same recognition. By 2022, Tata Steel had become India's largest steel producer by volume.³ Consolidated revenues stood at INR 2.4 trillion [\$32 billion^{b,4}], EBITDA margins were at 26%, up from 9% in 2013, and net debt to equity was down to 0.5. Tata Steel's stock price had outperformed the benchmark NIFTY Metal index over the last five years (see **Exhibit 1**). An analyst observed, "Aggressive. Assertive. Confident. After a decade, the business has a veritable spring in its step."⁵

With much of the heavy lifting behind him, Narendran turned his attention to the firm's B2C business. Since the early 2000s, Tata Steel had moved from entirely being B2B focused to earning a small percentage of revenues from retail consumers. In 2018, it accelerated its B2C drive by launching an e-commerce platform, *Aashiyana* ("Home" in the Hindi language), to sell products directly to retail clients. Aashiyana's offerings had grown from solid steel bars to Tata Steel home construction products (doors, windows, furniture) and services (design, service providers).

Now, Narendran and his team debated whether to open Aashiyana up to non-Tata Group third-party providers of ancillary home construction products, such as cement, paint, tiles, and home fittings. Some argued that this would create a unique one-stop shop for construction-related goods and could

^a The Global Lighthouse Network showcased manufacturing firms using Industry 4.0 to transform factories, value chains and business models, and generate financial and operational returns. As of 2022, the list included 90 firms, two of which were Indian.

^b USD-INR exchange rate of 75.9 as of March 2022

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enhance customer loyalty, but others felt that it was an unknown territory that would make the firm lose its focus.

The Indian Steel Industry

India was the world's second-largest steel producer, but with a 5.9% market share, it was small relative to China, which supplied more than 50% of the overall global supply (see **Exhibits 2-3**). In India, the construction sector was the most important consumer of steel (see **Exhibits 4-5**)⁶, and steel consumption per capita was 64.2 kilograms per person versus a global average of 229 kilograms, so there was tremendous scope for consumption to grow in the domestic market.⁷

While Indian steelmakers had access to a low-cost workforce and local iron ore reserves, they grappled with other challenges, including significant logistic costs, since most Indian steel plants were inland (see **Exhibit 6**). Other challenges included unreliable power, inadequate and unaffordable finance, and limited coking coal supplies.⁸ Consequently, a steel company in India that bought raw materials at market prices could have a delivered cost to the market of \$80-\$100 higher than other markets (see **Exhibit 7**). Indian steelmakers also lagged on technological know-how, with high value-added steel representing a marginal 8% of finished steel versus a global average of 20%.⁹

Steel manufacturers were also facing global headwinds. The steel industry was the world's second-biggest energy consumer, and manufacturers were under pressure to cut emissions. Indian steel producers lagged leading global peers on CO₂ emissions by 30% per ton of crude steel. Digital disruption was another major challenge. Industry 4.0, consisting of technologies ranging from the Internet of Things to drones and robots, offered solutions to many issues in the steel production chain, but adopting these technologies implied higher costs.¹⁰

Tata Steel Limited

Tata Steel was originally conceived by Jamsetji Tata, founder of the conglomerate Tata Group, in 1867. Unfortunately, he passed away before his vision could be realized. His eldest son, Sir Dorabji Tata, took forward his mission, establishing India's first steel company, "Tata Iron and Steel Company (TISCO)," in 1907. He set up a plant in a small town in the present-day state of Jharkhand in eastern India. Later, the town was renamed "Jamshedpur" in memory of Jamsetji.¹¹

Tata Steel's history was intertwined with that of modern India. The firm was ahead of its time in following labor best practices, including 8-hour workdays, medical leave, bonuses, provident fund, and skill development programs. It developed Jamshedpur into a model town,¹² embodying Jamsetji's belief that "in a free enterprise, the community is not just another stakeholder but is, in fact, the very purpose of its existence."¹³ Tata Steel's long history of giving back to the community was highlighted in its 1989 campaign through the slogan, "We also make steel."¹⁴

Over the years, the company expanded its operations in India and globally. In 2007, Tata Steel acquired the Corus Group that had plants in the UK and Netherlands.¹⁵ When Narendran took over as the firm's CEO in 2013, it was one of the world's few fully-integrated steel companies, with expertise in mining, manufacturing, and marketing finished products with plants in Europe, India, and Southeast Asia, two captive coal mines, and four iron ore mines in Jharkhand and Odisha.^{16,17}

Narendran's Tenure

Narendran was a Tata Steel “lifer,” having started his career at the firm in 1988 after completing his MBA from the prestigious Indian Institute of Management (IIM), Calcutta, and an undergraduate degree in engineering. He worked across operations, marketing and sales, the CEO's office, and foreign operations before becoming Tata Steel's CEO on November 1, 2013.¹⁸

Narendran took charge at a difficult time. Global steel demand had collapsed, leading to a secular price decline (see **Exhibit 8**).¹⁹ The European debt crisis followed, jeopardizing Tata Steel's European business. The firm had a net debt-to-equity ratio of about 1.8 in FY14.^c Simultaneously, the firm's iron-ore mines had closed for the first time in its 100-year history. Importing iron ore at a significantly higher cost subsequently put the firm under financial pressure.

Beyond the financial crunch, cultural change was also the need of the hour. Narendran had inherited a workforce that was comfortable and set in its way of doing things. Narendran ruminated, “When I took over, I was the youngest MD of Tata Steel. If, at 48, I'm the youngest, this is a very traditional company. How do you pivot a company with such a rich history to become more forward-looking when most employees are lifers?” Though the firm had modernized in bits, its operational efficiency was lagging. Narendran continued, “We were seen as a strong continuous improvement company, not an innovative company. Someone once told me, ‘We are the best public sector company to work for in India.’ We were respected for how we dealt with the community and employees. But we were not viewed as a leader in technology.”

Narendran viewed his age as a factor in his favor. “Assuming I could stay in this role till retirement, I had 17 years ahead of me and could think about the long-term. I wanted to make Tata Steel structurally, culturally, and financially future-ready. We needed to consider what elements of the culture we wanted to preserve, what we wanted to jettison, and what we wanted to create.”

As he embarked on a business transformation drive, Narendran had a solid partner in the firm's Chief Financial Officer (CFO), Koushik Chatterjee, who was also a Tata lifer. Having worked in various roles at Tata Steel and parent group Tata Sons, he was appointed group CFO in 2004.²⁰ Chatterjee reflected on their relationship, “Narendran and I have a strong bond. All our strategic decisions have one goal: ‘Is this the right thing for Tata Steel?’ When we hand over the baton to the next generation, we want to look back and be sure we ran the race as best as we could. We must ensure that Tata Steel is around for the next 100 years and stays alert, agile, and best-in-class.”

Business Transformation

Narendran conceived of various levers through which to transform the business. This included measures to deleverage the firm and to use the resulting free cash flow to invest in opportunities in India to cater to a strong and growing domestic demand base. Another was to venture into new materials. Safety, sustainability, and agility were three further goals, and finally, digitalization.

Deleveraging & Growth through Acquisitions

When Narendran took over, Tata Steel's debt levels were high. The biggest challenge was downsizing the European business, where high costs and the product mix were not viable in a low-growth environment. As a result, cost rationalization was critical. Narendran introduced cost-cutting

^c India's Fiscal Year (FY) ran from April 1st – March 31st.

measures, including personnel reduction, product mix and raw material changes, indigenization of spare parts used in maintenance, and improvements to operational efficiency, helping the European operations become self-sufficient. By FY22, the overall debt-to-equity had declined to 0.5.

The improved EBITDA margins, however, didn't lead to improved cash flows, a priority for Natarajan Chandrasekaran, Chairman of Tata Group since 2017. Narendran explained, "Coming in from the group's software arm, Tata Consultancy Services (TCS), Chandra had much less appetite for capital expenditure. He asked us, 'Where is the EBITDA going? I don't see any cash.'"

"Chandra's directive forced us to shift from a focus on the next big capital project to looking at what we needed to constantly do to hit the targets: lower fixed and variable costs, improve the supply chain, change the product mix, decide what we sell, to whom we sell and how we sell, and inventory and spares management," Chatterjee explained. "We need to ensure that margins stay at a certain level. Input price inflation is 5%-7% per year; we need to improve continuously to cover these costs."

Deleveraging created scope for Tata Steel to invest in buying out bankrupt firms that represented a once-in-a-lifetime opportunity for acquiring good assets at low prices. In May 2018, Tata Steel acquired Bhushan Steel, a bankrupt steel producer that was an attractive target with good asset quality, a strong product portfolio, proximate synergies, and alignment with Tata Steel's long-term strategy. Subsequently, Tata Steel bought some other smaller firms as well. Chatterjee reflected, "[We] will continue to examine such entities, but also be very disciplined in our capital allocation. We must ensure that our financial strength and value are protected and enhanced."

New Materials

With technological advancements, consumers were demanding materials that were stronger yet lighter than steel, quick to produce yet more durable, and economically and environmentally viable. Narendran set a goal for the firm to earn 10% of its revenues from new materials by 2025. He appointed Debashish Bhattacharjee as vice president of the technology and new materials business. Bhattacharjee had worked within the firm's R&D arm since completing his Ph.D. in material science and metallurgy from the University of Cambridge and had authored over 50 international peer-reviewed journal publications and filed 20 patents.²¹

Bhattacharjee took a structured approach to identify viable options for Tata Steel's foray into new materials. The new materials needed to have modest capital requirements, their production needed to leverage pre-existing firm processes or relationships, and they needed to create value in adjacent businesses. He shortlisted three verticals: fiber-reinforced polymers (FRP),^d graphene, and advanced ceramics. Bhattacharjee reflected, "New materials will ride megatrends; their consumption will only increase," adding, "Tata Steel has taken a disruptive leap. . . We are creating tomorrow today!"²²

Sustainability

Recognizing that Tata Steel needed to keep up with global developments in sustainability, Narendran appointed Sanjeev Paul, a senior operations employee, as the VP of safety, health, and sustainability. He explained, "Since Paul had run plants and subsidiary companies, he had the respect of employees. If I brought in someone who had never worked in the business, there was a risk they would bring in theory without the practical knowledge of how to apply it."

^d FRPs were stronger and lighter than metals, non-corrosive, could be quickly produced, and increasingly, recyclable.

The next step was educating senior leaders on current sustainability themes. The firm sent senior executives to the Cambridge Institute of Sustainability and brought in professors to educate them. Narendran reflected, “This exposure made us realize that sustainability is an existential issue. We have adopted advanced technologies to lower our carbon footprint, but moving toward zero carbon emissions remains complex and costly.”

Agility

Narendran explained, “External facing parts of the business automatically adapt due to market pressure, but agility is a problem in internal facing parts of any organization in a traditional sector.”

Narendran launched an agility project with the help of McKinsey. First, the leadership team was brought on board, and then people down the line. Measurable goals were set and built into performance management and review systems. When people retired, the right person was put in the right job, someone who could drive change. Narendran mused, “Once you establish the correct framework, it becomes a habit.” A “Bureaucracy Busters” team eliminated pockets of bureaucracy. The firm introduced a 360-degree feedback program that measured employees’ responsiveness, people development skills, accountability, collaboration, and a ‘work from anywhere’ policy.

Digital Transformation

Digital transformation lay at the core of the firm’s business transformation program. Narendran conceived the idea on his 2014 visit to the World Economic Forum, Davos, where digital and technology were the two big themes. At the time, at Tata Steel, “digital was seen as the IT guy’s job.” In 2016, he launched a digital transformation drive to improve efficiency and performance.²³

Bringing in Jayanta

Narendran knew hiring the right person to drive the digital transformation initiative was critical. He recollected, “I knew we must look for external talent. But joining Tata Steel at a senior level is not easy. We are a cradle-to-grave kind of firm.”

He zeroed in on Jayanta Banerjee, a 24-year veteran and digital transformation expert from TCS, and appointed him chief information officer (CIO) in February 2018. Narendran reflected, “I felt Jayanta had the maturity and capability to fit into Tata Steel at a senior level and the EQ required to drive transformation. I insisted he relocated to Jamshedpur.” Banerjee, who was settled in Mumbai with his family, moved to Jamshedpur for three years. He stayed in a bungalow in the Tata colony, attended all the social functions, and traveled to the plant sites. “I understood the people and their lifestyles, including those working in the plants located in dense forests.” Chatterjee added, “Jayanta brought a systematic approach to digitalization.”

Narendran and Banerjee agreed that any significant transformation could only occur if it were business-led. Narendran explained, “Jayanta’s job was to install the infrastructure, educate us, push projects, do whatever it takes, but the transformation had to be owned by the business.” Banerjee concurred, “Business has to adopt technology, run it, use it, make money from it; only then can technology be useful. IT processes are a craft; building the right integrated systems can be extremely complex. But business doesn’t care about that. Business needs simple outputs and simple solutions.”

Getting Buy-In: Building a Digital Mindset

A 2015 internal survey had previously highlighted two critical and persistent problems. First, senior managers were digital novices who were often non-committal and even resistant to digital implementation. Second, many employees were unaware of the potential applications of digital technologies and their benefits.²⁴

Banerjee knew that he had to get the buy-in of the Tata Steel leadership team. He began by putting in place a reverse mentoring program. Sixteen high potentials under 30 were chosen to mentor senior leaders, including Narendran, to introduce them to basic digital concepts such as search engine optimization, augmented reality, and data analytics that led to many “Oh! I didn’t know this!” moments for the senior leaders. The program was expanded the following year to pair more mentors with leaders. Banerjee also organized international road trips to expose senior executives to best practices in manufacturing digitalization and gain a greater appreciation for Industry 4.0 use cases. Narendran said, “People came back from these trips with a sense of ‘wow, a lot is happening that I was unaware of.’” Banerjee also invited experts from the top consulting firms to give presentations on leading digital applications.²⁵

Next, Banerjee’s team rolled out programs to create capabilities across the firm. Banerjee explained, “We set up a ‘digital value acceleration’ team that acted as catalysts or evangelists. They would go around the firm to explain analytics. The goal was to create a belief system.” The team also launched a mandatory program in which conversational video modules deconstructed key concepts.

Upgrading Technology & Creating Value²⁶

With educational projects underway, Banerjee began upgrading the firm’s archaic technology, augmenting its IT network, cloud, computing, and cyber security capabilities. Narendran explained, “Globally, firms were spending 2% of revenues on digital; at the time, we were at 0.3%. We needed to get to at least 1%.” Banerjee added, “The idea is to build the road before the car. It sounds trivial, but we did not even have WiFi in our mines and plants.” With widespread Internet connectivity, Banerjee led the migration of data and computing processes to the cloud. “Cloud brings agility,” Banerjee stated emphatically, “It means the data comes to you instead of you going to the data.”

At TCS, Banerjee had observed that only 30% of digital initiatives globally delivered sustained improvements. He knew that success would hinge a lot on the choice of initial projects to work on. His next step, therefore, was to use a hybrid approach to assess where technology could create business value. Ideas could be bottom-up, stemming from a business need; top-down, driven by a strategic goal; or outside-in, inspired by best practices at competitors. Narendran explained, “It is not only about putting your foot on the accelerator (top-down approach); the car should move quickly even when you lift your foot off the accelerator. For that, you need the power of bottom-up. Every employee should feel connected with the firm’s vision.”

For bottom-up ideas, business teams had three directives: to create or re-imagine business models to leverage digital, identify disruptive opportunities and threats posed by digital, and build richer experiences for key stakeholders. The target was to “*think big, start small, scale fast.*” On thinking big, teams were encouraged to benchmark themselves against the global best practices of digital frontrunners. A key feature of the benchmarking exercise was that it was process-specific. On maintenance, it analyzed best practices at firms like GE and Rolls Royce in the airline industry. For data analytics, it looked at cutting-edge firms in operations. For HR, it looked at firms like Unilever, known for best-in-class recruitment and retention programs.

For top-down driven ideas, leaders were asked to formulate clear vision statements of how their organization could adopt digital over five years and to cancel pet projects if they did not align with the firm's top priorities. To encourage digital adoption, performance metrics were balanced between short-term performance and long-term improvements. A monthly committee was set up for leaders to discuss technology adoption across the value chain.

Banerjee was given a pool of funds to invest in chosen projects, with final approval vesting with Narendran. He applied two criteria to assess projects. The first was that the project had to generate significant financial value given that he had an overall target of generating \$2 billion in EBITDA savings and an expectation of a 6-8X return on funds invested. The second was that it should significantly impact the stakeholder experience. Banerjee explained, "Digitalization can offer intangible benefits as well, for example, by improving safety, that can create a long-term competitive advantage."

The team faced pockets of resistance as they rolled out the program. For example, the cost pressures being faced by the firm led some to question whether spending on the digitalization drive was necessary. Chatterjee stated, "People complained that I hadn't funded their requests for investment in operating assets, but I was giving a lot of money to Jayanta. They said we are a steel company; we should invest in steel assets rather than digital ones." Further, divisions running traditional cost-saving programs felt marginalized, fearing digital cost-saving projects would take the limelight. Narendran assuaged their concerns by acknowledging everybody's contributions. "Rather than waste time figuring out who contributed how much to the savings, I said, as long as Tata Steel benefits, let everyone take the credit and share it. The key was to build momentum and keep things moving."

The firm categorized digitalization projects into five stages, D0 to D4. D0 projects were those at the ideation stage, and D4 was at the completed stage. Narendran explained, "We religiously reviewed progress towards targets in every monthly meeting. If I saw divisions with consistently few D0 projects, I reminded them they had a weak pipeline. I had to build momentum by constantly talking about digital, pushing it, explaining its relevance, and complimenting teams where I saw traction. Digital was part of every review." Bhattacharjee dwelt on Narendran's involvement, "Narendran's persistence in tracking targets pressured the organization to change. No initiative to create cultural change can be successful without alignment, and cultural change has to be driven from the top-down."

Digitalizing Manufacturing Processes and Operations

Banerjee's team worked extensively with business teams across the production chain to find ways to "digitalize" processes. His team worked with business operations to help them identify problems. Banerjee explained, "Change does not happen through technology; it happens through mutual trust and belief. We needed to understand the business context, translate pain points into IT language, and find solutions. If the context is not understood, projects will fail."

One such important initiative was using technology and analytics to eliminate product defects. Bhattacharjee explained, "Our product quality was not at par with global leaders like Nippon Steel. Quality can be improved either by inspecting out more defects or through better process control." Traditionally, Tata Steel posthumously inspected each coil, rejecting defects as they were found.

A review of the process revealed that over 400 interdependent parameters determined quality in the production process, and there was only so much manual inspection and intervention possible in such a value chain. Working with the operations team, Banerjee's team developed a digital "surface inspection system." They installed sensors along the length of the production chain. AI and machine learning analyzed data collected from these sensors, identifying production patterns for every steel grade and deviations from the standard trend, i.e., the defects.

As this technology was installed, some workers felt threatened by the change. Banerjee's team spent a lot of time with these workers, explaining how technology could help them do their jobs better rather than take them away. He explained, "The operator who used to look at the sheets through the naked eye and make a judgment call is now equipped with a high-definition surface inspection system, calling out the cracks or defects early on and making better decisions. The operator still takes the call, but the error count has reduced. We call it 'assisted AI.'"

There were countless such examples. Narendran shared, "There used to be a guy counting the number of pipes coming out of our tube mill. With technology, a scanned photograph can count the number of pipes in a second." High-definition cameras and other sensors ensured that operators could assess flame temperature and other parameters of a blast furnace safely from afar rather than standing next to the furnace and risking being injured by a flare.

As workers saw the many benefits of technology, their mindset changed. Banerjee elaborated, "Workers started feeling safe as they saw how technology prevented accidents. They appreciated how technology made their jobs easier. This made the union workers ask for more digital enablement." Banerjee reflected, "It starts with an IT mandate, but we eventually want the business to own it. We want businesses to call my team leaders saying, 'I want this dashboard or analytics or use case.' You can't put a digital scientist in the steel plant and hope things will improve. A process engineer must work with a digital scientist to achieve the correct application of digitalization. Only when you understand the cause of the defect that you can eliminate it."

Other tools helped improve asset efficiency. From conversations with business teams, Banerjee realized that predicting asset failure was a major pain point. His team developed an algorithm to predict equipment failure and prioritize assets for maintenance. Technology also helped put assets on "auto-pilot." Narendran elaborated, "Operating a blast furnace has become a science. Using data, we can minimize variability between shifts. We can also simulate new technology online, minimizing the time that assets have to be taken offline for testing."

Over time, as technology became embedded across the firm's operations, Banerjee's team initiated a data office initiative to consolidate data residing in various silos across Tata Steel's plants, mines, and supply chain, making it accessible to teams across divisions, anywhere, to enable "smarter" decision-making. Narendran explained:

We employ thousands of contract workers. Due to technology, we now know exactly how many are coming in and how much time they spend in the plant. We can improve productivity by managing these workers' hours, using them across multiple teams and jobs more efficiently. We developed a watch along with Titan (a Tata Group company) to improve the safety of workers operating cranes and in other dangerous roles. If a worker's vitals change, the technology immediately alerts his supervisor.

As systems moved online, the workforce needed to adapt to doing things digitally. Some senior managers were reluctant. Vice President Human Resource Management Atrayee Sanyal mused:

We digitalized certain tasks, and employees were not given a choice. Some people resisted changing habits. They were used to using paper and pen or someone else generating the report. Suddenly, much of this got digitalized. Now, they had to do it themselves and on the digital system. Many of our employees were not used to this. They had spent decades working in the analog world. When you introduce change, initially, there is chaos, but people settle down if we show a firm intent to change.

Another example was the digitalization of supply chain processes. Previously, the firm would often face a backlog of orders for a particular product in a local market while there would be an excess inventory of the same product in another market - a result of inefficient forecasting systems that could not anticipate changes in local demand and were unresponsive to variations in the field. Under the new program, analytics were introduced to forecast local demand, assess which customers to serve, determine where to manufacture and store products, and reduce raw material and inventory costs. An executive explained, "Planning demand on a granular level allows us to allocate assets and capacities accurately. We use analytics to assess if demand is falling short in a local market and find alternative customers to take the volume. Rather than discount to move such materials, we now are able to maintain pricing discipline."²⁷

Managing Change & Next Steps

Narendran reflected, "In any change program, some people start running straight away, while others take time to understand the potential. Marketing and sales were quicker to adapt than operations and engineering. The naysayers need to be pushed. Digitalization is a journey, not a destination."

One strategy used by the leadership was to showcase the success of early adopters. Banerjee reflected, "It was important to let business leaders take credit. I told my team to bask in the glory of their success." Chatterjee agreed, "Recognition served as a big motivation." Bhattacharjee highlighted Banerjee's role in driving change, "When Jayanta came, there was skepticism. But he was able to break through barriers and implement projects." Narendran agreed, "Jayanta fit in wonderfully; he built a lot of equity and pushed hard without getting caught in the crossfire. He brought in domain knowledge, experience, and long-term thinking, changing the language and vocabulary."

Ultimately, the firm's fabric and workforce also needed to change. Narendran said, "When people retire, new blood comes in; this builds momentum." Bhattacharjee shared:

As long as the leadership understands the need for entrepreneurial people who can fail fast and move on, and supports their hiring, it will give our new ventures sufficient incubation space. Hiring open-minded workers willing to learn new ways is more important than toppers set in their way of doing things. Workers need to continuously reskill to adapt to change. The skills needed today will not be the same ones required tomorrow. We may destroy the company by preserving people unwilling to adapt; it will lose relevance. An excellent manufacturing firm is one with excellent internal process controls, excellent human interfaces, and excellent delivery. The human element, having the right kind of employees, is critical.

By 2022, the digitalization program was self-sustainable, with business teams taking ownership of the project and value realization.²⁸ Banerjee noted, "Business leaders are now 100% believers. We have over 850 digitalization projects in place. I don't have to push them; they push me. We cannot imagine running any process without a digital layer."

Tata Steel's next goal was to digitalize all critical processes. It aspired to create virtual command centers that allowed managers to monitor and correct plant functioning remotely. "Shell and Rio Tinto manage oil rigs in the middle of the ocean from afar. So why can't we run our mines in Noamundi (200 km from Jamshedpur, where Tata Steel's captive mines were located) from Jamshedpur?" mused Narendran.

Digitalization at the Customer End

The Foundations of the B2C Business

Since the late 1990s, Tata Steel had been working on building a B2C business to counterbalance the ups and downs of demand from its B2B customers. Narendran reflected, “We had great relationships with construction firms and contractors but none with homeowners who typically bought steel rebars, tubes, sheets or plates, and wire products from retail hardware stores to construct roofs, gates, garages, and internal foundational structures for their homes”.

Narendran explained, “The B2C business offers better margins than B2B. B2C clients are typically one-time buyers and are less price sensitive than B2B buyers. They just want a few tons of trustworthy stuff. Usually, they depend on advice from people they trust in making their decisions.” Tata Steel involved key influencers in the home-building process to boost sales. Gupta explained:

While building a house, two people come into one’s life, an architect and a contractor. Both can make or break a brand. We started to work on both of them. The third influencer is a satisfied customer who might not buy more from us but will enthusiastically recommend us to their friends and family. When prospective customers walk into a shop, we want them to say, ‘I want Tata Steel.’

To improve brand recognition, Tata Steel started packaging its products. Gupta reflected, “Historically, we had no wrappers, packaging, or price. We started to work on these fronts to create our brand differentiation.” Gupta expanded:

Steel was sold traditionally by the kilogram, and customers did not know if they were being cheated. Further, architects quoted steel requirements in meters. We started selling steel by the piece, resolving the issue of customers having to weigh the steel to ensure they had not been defrauded. The change was powerful.

Tata Steel realized that success in the B2C business hinged on creating links with the hardware stores where the retail customer shopped for steel products. Since 2001, it had been developing distributors who could supply its products to these stores. By 2014, it had 2-3 such distributors in each state, through which it cumulatively had links with 10,000 hardware stores across the country. “This is the largest such distribution network in the country in our sector,” noted Peeyush Gupta, VP of supply chain, an engineer and an MBA by qualification who had joined the firm more than 28 years ago and worked across the firm’s marketing, sales, strategy, and finance divisions. He continued, “Competitors can copy our packaging, branding, or pricing strategy, but creating a retail distribution engine in India is difficult.” Gupta further explained the historical evolution:

We worked with our channel partners to develop win-win business relationships. Tata Steel has been associated with most of its distributors for decades and across generations. Tata Steel’s credibility leads to an unparalleled loyalty that is not shaken by transactional headwinds. Our loyal distributors regularly and consistently buy from us and, in turn, give exacting levels of service to their retail dealers.

We have endeavored to give distributors a 16%-24% annual return on invested capital. This channel also stands to benefit from Tata Steel’s B2C growth. As our business grows, channel revenues will grow, and their family-run companies will expand. Retail customers want price stability and confidence in weightment. They detest the retailers making a mockery of the list price.

We introduced the concept of “right price, right place, and right weight” through our retail sales channel, thereby assuring consumers they were getting the right price. For weight assurance, we started to “sell by piece” to address customers’ weight mismatch concerns. The place assurance was given by calling retail stores “authorized by Tata Steel.” The stoppage of inter-retail store rivalry was perhaps the single most significant reason for the channel discipline and consumer delight we have created.

Our unique 2-tier distribution model of distributor-retailer has enabled a pan-India coverage that is hard for other firms to match.

Aashiyana

In May 2018, Tata Steel launched Aashiyana (“house”), an e-commerce platform to sell steel rebars. An analyst noted, “Tata wanted to sell steel the way other packaged products are sold in India: in smaller amounts, directly to the consumer. This had never been done before.”²⁹ While there was unanimous agreement among analysts that Aashiyana created value for customers, there were some concerns raised about loyal, existing Tata Steel distributors being disintermediated.

Gupta clarified, “Aashiyana was about enabling customers to purchase steel bars online and assuring delivery within a certain timeframe. It was because of our existing distribution channel that we could guarantee delivery within 72 hours of purchase anywhere in the country. We cannot ship a few pieces per order from Jamshedpur by ourselves (steel plant location). We need our distributors to be able to deliver on our promise to the B2C customers. Our distributors are accountable for checking the online platform and fulfilling customer orders within 72 hours. If they miss the deadline, they will lose business. But if they do things right, Tata Steel will not break their trust.”

Narendran added, “Now, anyone in India or overseas can order steel to construct a house anywhere in the country. The order generation process can happen anytime, anywhere.” Gupta added, “It is interesting to note that the average purchase offline is about 1.6 tons per customer, but on Aashiyana, it is 2.7 tons per customer. Revenues from Aashiyana had grown to INR 14.7 billion in FY22, up 100% from the previous year. While sales from Aashiyana were still relatively small, it helped Tata Steel generate new customer insights and retain a longer touch time with consumers through their home-building journey (typically 12-18 months), thus adding to brand equity and recall.”³⁰

Tata Steel also invested in systems to service digital orders that minimized logistics, warehousing, and handling costs for its distributors. It pushed distributors to invest these savings in bigger warehouses and better technology for their sales reps. Gupta elaborated, “An opaque channel has now become fully transparent. We want our distributors to invest more and provide unmatched service to customers anywhere in the country within the promised timeline.”

However, building out the B2C business required a different skillset and mindset. Narendran said, “It’s making sure that the door we promise reaches the customer on the day of their housewarming ceremony. We get a lot more exposure on social media, and it keeps us on our toes. We are dealing with customer-related and customer-initiated issues that we hadn’t dealt with before.” Banerjee described, “With technology, internal systems can be complex, but the customer interface should be simple and their experience flawless.”

The Dilemma

As Aashiyana's revenues grew, the firm sought additional ways to create value. Gupta said, "Aashiyana was initially focused on steel bars. But customers building homes need other products and services. We wanted to offer these. Once a customer came on our website, we wanted them to stay on it."

Tata Steel started selling its other existing steel products on Aashiyana. This shift required some adjustments, as these products were hitherto sold and run under different teams within the firm, but on Aashiyana's platform, they needed to be available to the customer in a consistent and seamless format.³¹ Seeing a gap in the market, Tata Steel also ventured downstream into the manufacturing and selling of steel doors, windows, and furniture. Finally, Aashiyana began to offer services such as designs, estimates, and new material ideas and listed key service providers by location.

The firm also planned to offer other Tata Group companies' construction-related products, such as Voltas air conditioners, Tata Capital loans, Tata AIG insurance, Croma home appliances, and Tata Sky cable TV connection, on the platform. Once Tata Steel acquired the customer, it had visibility over customer needs and could cross-sell Tata Group products. Some leaders wanted to go even further and convert Aashiyana into a curated online shop for all construction goods. The idea was to offer ancillary home construction products from select trusted non-Tata Group providers on the platform from whom Tata Steel would receive a commission on each sale. An executive elaborated on this view:

We can enhance the customer experience and organize the home construction market. Partner companies must agree to specific terms such as no price gouging, no bait and switch, on-time delivery, and price transparency. We may have to do additional developmental work as we did for steel, but it would be worth it. Currently, we touch only 25% of the customer's wallet. Can we leverage our platform to reach 100% of the customer's wallet and own the customer journey?³²

Yet another proponent took the idea further and reasoned, "If we don't do this, somebody else will. We should set it up as an independent venture, create a focused team, and let them build it from scratch. They will be responsible for selecting and developing partners, curating the list of products for the website, and managing the entire ecosystem. For that team, this business will be everything."

Others were cautious, noting that opening the platform to outside firms was risky. Gupta explained:

With Tata Group companies, we have some soft control; we can call them to ensure the customer order is fulfilled on time. With external non-group companies, there might be a limited sense of loyalty and commitment to our platform. We also have no visibility into their supply chain. If a customer has a bad experience, how will it affect our brand? We want to open our platform, but what if we lose control of the customer experience? Today, we guarantee customers that the price on the platform for all Tata Group products is the same as in physical stores. With third parties, we will not have this kind of control or the ability to provide the same level of consistency.

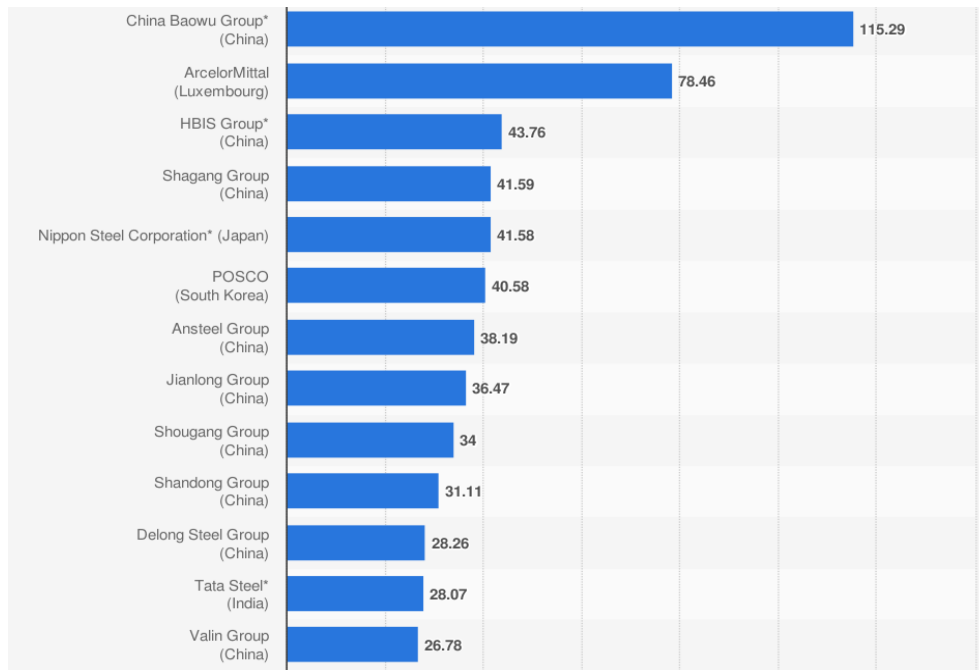
These executives also felt that the threat of a competitor launching such an endeavor was low, given Tata Steel's strong position in the steel market. One explained, "If you are Amazon or any other e-commerce competitor, we will not sell our products to or through you. Retail customers will have to purchase directly from us. Manufacturers list on Amazon to increase sales, but we don't need more sales. We are selling directly to half a million B2C customers every year."

Narendran could see both sides of the argument. He was prepared to make the final decision and called his top team for one last brainstorming session.

Exhibit 1 Tata Steel's Stock Price versus the Benchmark Nifty Metal Index

Source: Google Finance, "Tata Steel Limited,"

https://www.google.com/finance/quote/TATASTEEL:NSE?hl=en&comparison=INDEXNSE%3ANIFTY_METAL&window=5Y, accessed April 2022

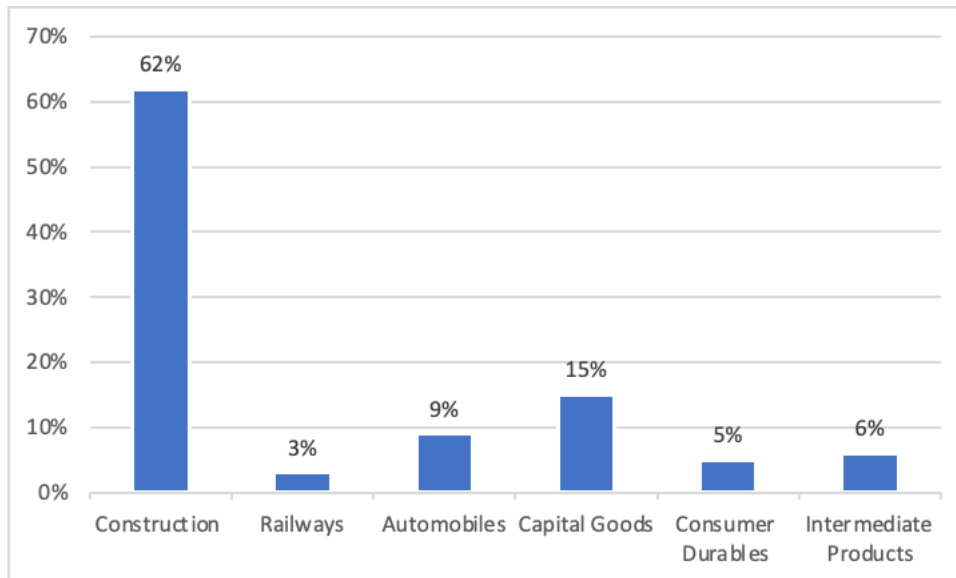
Exhibit 2 World's Largest Crude Steel Producers in 2020 by Production Volume (in MT)

Source: World's Largest Crude Steel Producers by Volume, via Statista, accessed September 2022

Exhibit 3 India's Top Steel Producers

Name	Description	Capacity (MT p.a.)	Revenue (INR bn)
JSW Steel	Leading integrated steel manufacturer, and India's largest steel exporter, with a presence in over 100 countries. Plants in Karnataka, Tamil Nadu, and Maharashtra. Targets production of 40 MTPA within 10 years.	22	1,479
Steel Authority of India (SAIL)	Government-owned steel making company. One of India's seven Maharatnas. Produces iron and steel at five integrated plants and three special steel plants, located principally in the eastern and central regions of India, close to domestic sources of raw materials. Manufactures and sells a broad range of steel products.	21	1,035
Jindal Steel and Power	Part of the diversified Jindal Group. Operates some of India's most advanced steel manufacturing plants across Chattisgarh, Odisha, and Jharkhand. Has captive iron-ore mines in Odisha.	11	453
Essar Steel	Known for high quality of flat steel products. Focuses on high value-added grade steels. Places great emphasis on sustainability, and is aligned with the WSA sustainability indicators. It is a zero-waste company.	10	
Rashtriya Ispat Nigam Ltd. (RINL)	Has an exclusive product mix of long steel products, is the largest producer of "bars and rods" in the country.	7	280

Source: Casewriter, compiled with data from: Indian Companies, "Top 10 Steel Companies in India (Manufacturer)," <https://indiancompanies.in/top-companies-in-steel-industry-in-india-by-capacity-market-share/>; IBEF, "Industry – Steel – Showcase – JSW Steel," <https://www.ibef.org/industry/steel/showcase/jsw-steels>; OK Credit, "Manufacturers – Steel Manufacturers in India," <https://okcredit.in/blog/steel-manufacturers-in-india/>; accessed September 2022.

Exhibit 4 Sector-Wise Demand for Steel in India

Source: Casewriter, adapted from PWC India, "The Indian Steel Industry: Growth, Challenges, and Digital Disruption," November 2019, <https://www.pwc.in/assets/pdfs/consulting/technology/the-indian-steel-industry-growth-challenges-and-digital-disruption.pdf>, accessed September 2022.

Exhibit 5 Drivers of Indian Steel Consumption

Construction: This sector comprised physical infrastructure (excluding railways) and real estate. India was forecast to become the world's third largest construction market by 2025. Infrastructure spending, a focus area for the government, was expected to drive growth in this sector. Within real estate, affordable housing was expected to lead the way. India's urbanization rate was set to rise from 33% to 40% by 2030-31. This translated to 90 million people moving from rural to urban areas. Consequently, the demand for housing in urban and semi-urban areas was expected to grow significantly over the medium term.

Railways: This sector was growing fast, driven by projects like track renewal and addition of new lines, 100% track electrification, dedicated freight corridors connecting industrial hubs in western and eastern India, and high-speed rail corridors.

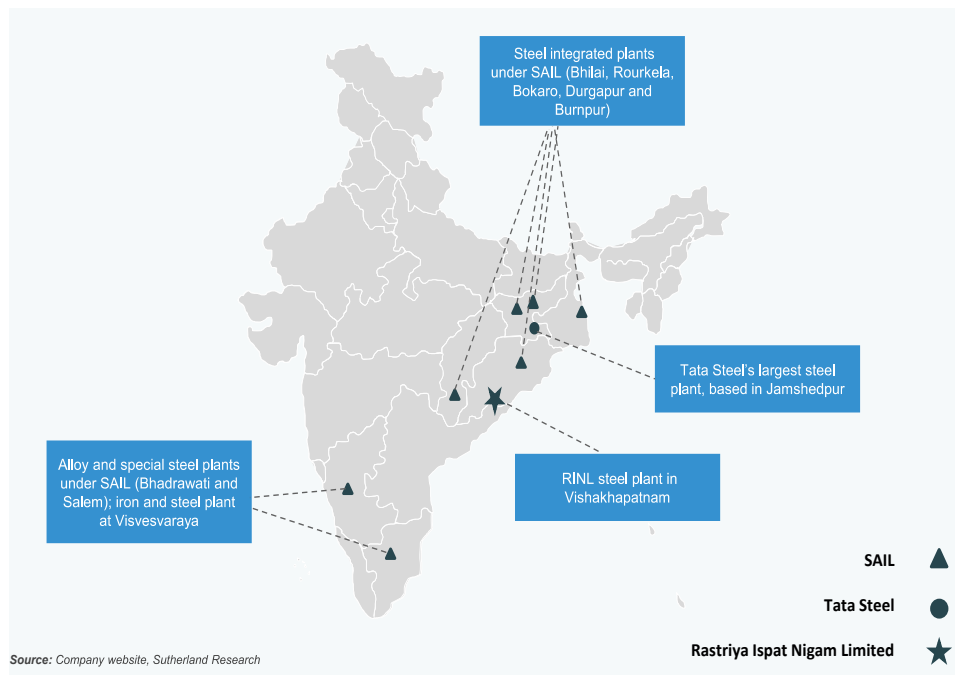
Automobiles: The Indian automobile industry was the world's fourth largest. It was focused on the domestic market, with domestic sales accounting for more than 80% of the total. The auto sector was expected to cross \$250 billion by 2026. The Government had formulated the Automotive Mission Plan 2016-26 which aimed at sustained growth in all sub-segments; this was expected to continue driving demand for steel. However, the plan also put significant focus on the manufacturing of electric vehicles which required less steel.

Capital Goods: In this segment, machinery and equipment were the two most prominent sub-sectors. This segment was dependent on construction, mining, and heavy and light industries. Thus, it was dependent on economic growth, and was therefore highly cyclical, with a wide variation in annual demand. It was also heavily dependent on imports, particularly in the large machinery segment. There was low acceptance of domestically manufactured products, with firms citing significant gaps in technological capabilities. Indian steel companies needed to invest more in R&D and technology to keep pace with more sophisticated foreign suppliers. The government had introduced initiatives to boost R&D spend and encourage technology providers and OEMs to set up base in India.

Consumer Durables: This segment was quite broad-based and included categories such as air conditioners, refrigerators, washing machines, and others. Given India's large young population, ongoing urbanization process, and increase in debt penetration, it was expected to see continued growth.

Intermediate Products: This segment was closely linked with the auto and oil and gas sector, besides industrial activity.

Source: Casewriter, excerpted from PWC India, "The Indian Steel Industry: Growth, Challenges, and Digital Disruption," November 2019, <https://www.pwc.in/assets/pdfs/consulting/technology/the-indian-steel-industry-growth-challenges-and-digital-disruption.pdf>, accessed September 2022.

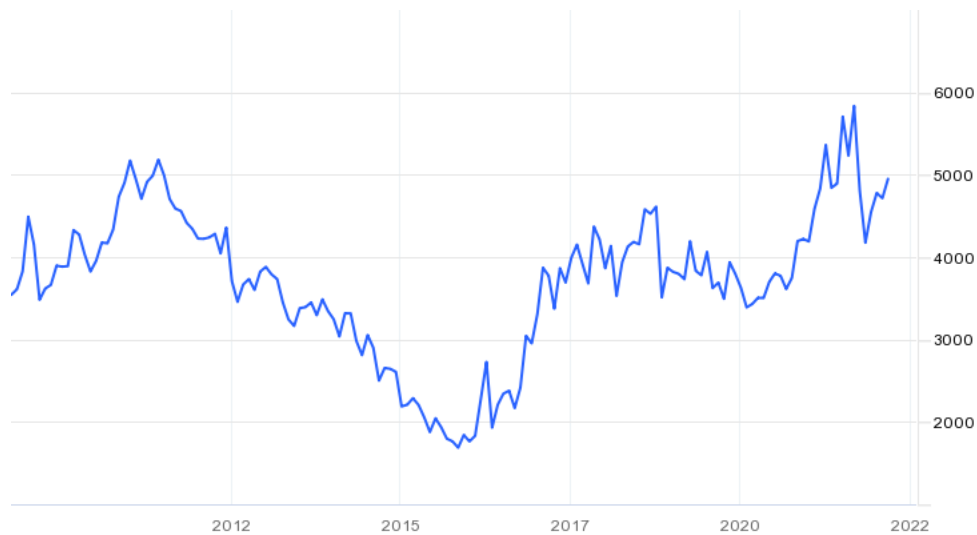
Exhibit 6 Location of India's Major Steel Plants

Source: India Brand Equity Foundation, "Steel," November 2021, <https://www.ibef.org/download/Steel-November-2021.pdf>, accessed September 2022.

Exhibit 7 Factors Reducing the Competitiveness of Indian Steel

(USD/tonne)	
Logistics and infrastructure	25–30
Power	8–12
Import duty on coal	5–7
Clean Energy Cess	2–4
Taxes and duties on iron ore	8–12
Finance	30–35
Total cost disadvantage	80–100

Source: PWC India, "The Indian Steel Industry: Growth, Challenges, and Digital Disruption," November 2019, <https://www.pwc.in/assets/pdfs/consulting/technology/the-indian-steel-industry-growth-challenges-and-digital-disruption.pdf>, accessed September 2022.

Exhibit 8 Steel Rebar Prices (CNY/T)

Source: Trading Economics, "Commodity - Steel," <https://tradingeconomics.com/commodity/steel>, accessed September 2022.

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